

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A&APEX) Date: 23-03-26
Time: 3 Hrs Max. Marks: 300

23-03-26_SR.IIT_STAR CO-SC(MODEL-A&APEX)_JEE MAIN_SPL GTM-12_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

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PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

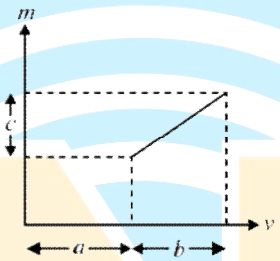
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. The energy of a system as a function of time t is given as $E(t) = A^2 \exp(-\alpha t)$, where $\alpha = 0.2 s^{-1}$. The measurement of A has an error of 1.25%. If the error in the measurement of time is 1.50%., the percentage error in the value of E $E(t)$ at $t = 5$ s is

A) 1 B) 2 C) 3 D) 4

2. The graph shows how the magnification m produced by a thin lens varies with image distance v. What is the focal length of the lens used?



A) $\frac{b^2 c}{a}$ B) $\frac{b^2}{ac}$ C) $\frac{a}{c}$ D) $\frac{b}{c}$

3. A particle is moving in a circular path of radius a under the action of an attractive force with potential energy $U = -k/2r^2$. Its total energy is

A) $-\frac{3}{2}\frac{k}{a^2}$ B) $-\frac{k}{4a^2}$ C) $\frac{k}{2a^2}$ D) *Zero*

4. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: If we move from poles to equator, the direction of acceleration due to gravity of earth always points towards the center of earth without any variation in its magnitude.

Reason R: At equator, the direction of acceleration due to the gravity is towards the center of earth. In the light of above statements, choose the correct answer from the options given below

A) Both A and R are true and R is the correct explanation of A.

B) Both A and R are true but R is NOT the correct explanation of A

C) A is true but R is false

D) A is false but R is true

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5. The ratio of surface tensions of mercury and water is given to be 7.5 while the ratio of their densities is 13.6. Their contact angles, with glass, are close to 135° and 0°, respectively. It is observed that mercury gets depressed by an amount h in a capillary tube of radius r_1 while water rises by the same amount h in a capillary tube of radius r_2. The ratio, (r_1/r_2), is then close to

A) 2/3 B) 3/5 C) 2/5 D) 4/5

6. The number of air molecules per cm³ increased from 3×10^{19} $to 12\times10^{19}$. The ratio of collision frequency of air molecules before and after the increase in number respectively is

A) 0.75 B) 0.50 C) 0.25 D) 1.25

7. If the electric potential at any point (x, y, z) m in space is given by $V=3x^2$ volt. The electric field at the point (1,0,3)m will be

A) $3\ Vm^{-1}$, directed along positive x-axis.

B) $3\ Vm^{-1}$, directed along negative x-axis.

C) $6\ Vm^{-1}$, directed along positive x-axis

D) $6\ Vm^{-1}$, directed along negative x-axis.

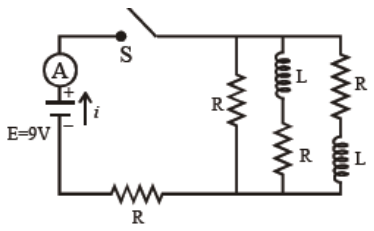
8. A current of 5 A passes through a copper conductor (resistivity $=1.7\times10^{-8}\Omega m$) of radius of cross-section 5 mm. Find the mobility of the charges if their drift velocity is $1.1\times10^{-3}\ m/s$

A) $1.3\ m^2/Vs$ B) $1.5\ m^2/Vs$ C) $1.8\ m^2/Vs$ D) $1.0\ m^2/Vs$

9. A capacitor C is fully charged with voltage V_0. After disconnecting the voltage source, it is connected in parallel with another uncharged capacitor of capacitance $\frac{C}{2}$. The energy loss in the process after the charge is distributed between the two capacitors is:

A) $\frac{1}{2}CV_0^2$ B) $\frac{1}{3}CV_0^2$ C) $\frac{1}{4}CV_0^2$ D) $\frac{1}{6}CV_0^2$

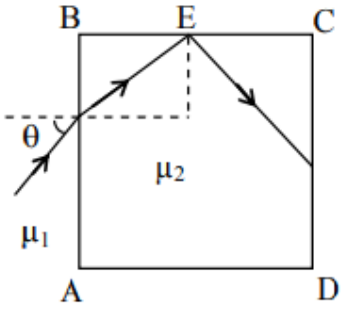
10. Figure shows a circuit that contains four identical resistors with resistance R = 2.0 Ω, two identical inductors with inductance L = 2.0 mH and an ideal battery with emf E = 9 V. The current 'i' just after the switch 'S' is closed will be



A) 2.25 A B) 3.0 A C) 3.37 A D) 9A

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11. A transparent cube of side d, made of a material of refractive index μ_2, is immersed in a liquid of refractive index $\mu_1 (\mu_1 < \mu_2)$. A ray is incident on the face AB at an angle θ (shown in the figure). Total internal reflection takes place at point E on the face BC. The θ must satisfy



A) $\theta < \tan^{-1}\frac{\mu_1}{\mu_2}$ B) $\theta < \sin^{-1}\sqrt{\frac{\mu_2^2}{\mu_1^2}-1}$ C) $\theta > \sin^{-1}\frac{\mu_1}{\mu_2}$ D) $\theta > \sin^{-1}\sqrt{\frac{\mu_2^2}{\mu_1^2}-1}$

12. A single slit of width a is illuminated by a monochromatic light of wavelength 600 nm. The value of 'a' for which first minimum appears at $\theta = 30^o$ on the screen will be:

A) 0.6μm B) 1.2μm C) 1.8μm D) 3μm

13. In photoelectric effect:

I. The photocurrent is proportional to the intensity of the incident radiation.

II. Maximum Kinetic energy with which photoelectrons are emitted depends on the intensity of incident light.

III. Max. K.E with which photoelectrons are emitted depends on the frequency of incident light.

IV. The emission of photoelectrons require a minimum threshold intensity of incident radiation.

V. Maximum kinetic energy of the photoelectrons is independent of the frequency of the incident light.

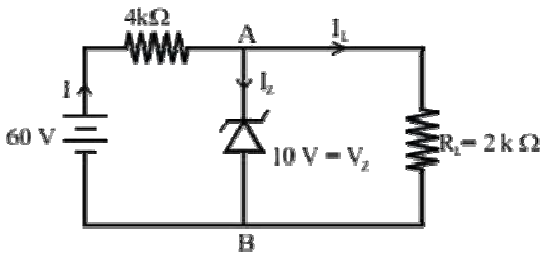
Choose the correct answer from the options given below:

A) I and V only B) I and III only

C) I and II only D) II and III only

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14. A Zener diode is connected to a battery and a load as shown below:



The currents, I, I_Z and I_L are respectively.

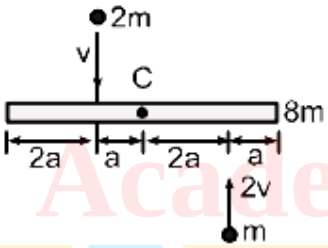
A) 15 mA, 5mA, 10 mA

B) 15mA, 7.5mA, 7.5mA

C) 12.5mA, 5mA, 7.5 mA

D) 12.5 mA, 7.5 mA, 5mA

15. A uniform bar of length 6a and mass 8m lies on a smooth horizontal table. Two point masses m and 2m moving in the same horizontal plane with speeds 2v and v respectively, strike the bar (as shown in figure) and stick to the bar after collision. The velocity of the centre of mass, angular velocity about centre of mass respectively are



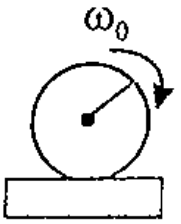
A) $0, \frac{V}{5a}$

B) $0, \frac{V}{6a}$

C) $0, \frac{V}{3a}$

D) $0, \frac{V}{8a}$

16. A uniform circular ring of radius R is first rotated about its horizontal axis with an angular velocity ω_o and then carefully placed on a rough horizontal surface as shown. The coefficient of friction between the surface and the ring is μ. Time after which its angular speed is reduced to 0.5 ω_o is:



A) $\frac{\omega_o \mu R}{2g}$

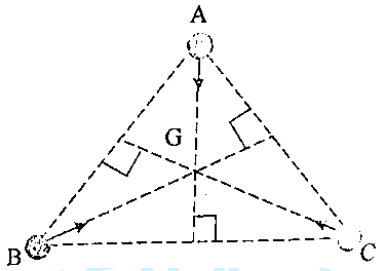
B) $\frac{\omega_o g}{2\mu R}$

C) $\frac{2\omega_o R}{\mu g}$

D) $\frac{\omega_o R}{2\mu g}$

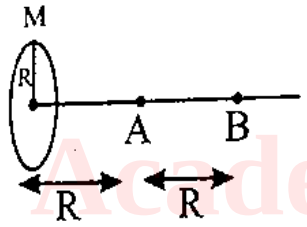
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17. Three particles A, B and C of equal masses, moving with the same speed 'v' along the medians of an equilateral triangle, collide at the centroid G of the triangle. After collision, A comes to rest and B retraces its path with a speed 'v'. The speed of C after the collision is:



A) v along BG B) v/2 along GB C) zero D) v along CG

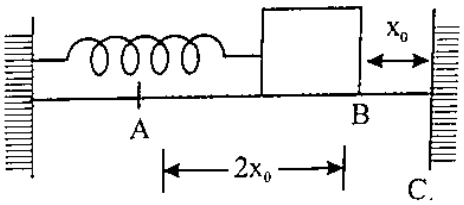
18. A ring (fixed) has a non-uniform distribution of mass, and its total mass is mass 'M' and it's radius is 'R'. A point mass m_o is moved from A to B along the axis of the ring slowly. The work done by external agent against the gravitational force of ring in moving the mass from A to B:



A) $\frac{GMm_o}{\sqrt{2}R}$ B) $\frac{GMm_o}{R}\left[\frac{1}{\sqrt{2}}-\frac{1}{\sqrt{5}}\right]$

C) $\frac{GMm_o}{R}\left[\frac{1}{\sqrt{5}}-\frac{1}{\sqrt{2}}\right]$ D) $\frac{GMm_o}{\sqrt{5}R}$

19. One end of a spring of force constant K is fixed to a vertical wall and the other to a body of mass m resting on a smooth horizontal surface. There is another wall at a distance x_0 from the body as shown in figure. The spring is then compressed by $2x_o$ and released. The time taken to strike the wall is:



A) $\frac{\pi}{6}\sqrt{\frac{m}{K}}$ B) $\sqrt{\frac{m}{K}}$ C) $\frac{2\pi}{3}\sqrt{\frac{m}{K}}$ D) $\frac{\pi}{4}\sqrt{\frac{m}{K}}$

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20. A rigid diatomic ideal gas undergoes a polytropic process at room temperature. The relation between temperature and volume for this process is $TV^2 = \text{constant}$, then molar specific heat capacity of the process is

A) 4R B) 2R C) R D) 3R

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. A tuning fork of frequency 340 Hz resonates with an air column of length 125 cm in a empty cylindrical tube closed at one end. When water is slowly poured in tube, the minimum height of water required for observing resonance once again is ___ cm. Neglect end correction.

(Velocity of sound in air is 340 ms^{-1})

22. 512 identical drops of mercury are charged to a potential of 2 V each. The drops are joined to form a single drop. The potential of this drop is _____ V.

23. A circular coil of radius 10 cm is placed in a uniform magnetic field of 3.0×10^{-5} T with its plane perpendicular to the field initially. It is rotated at constant angular speed about an axis along the diameter of coil and perpendicular to magnetic field so that it undergoes half of rotation in 0.2 s. The maximum value of EMF induced (in μV) in the coil will be close to the integer_____(assume $\pi^2 = 10$).

24. The electric field intensity produced by the radiation coming from a 100 W bulb at a distance of 3 m is E. The electric field intensity produced by the radiation coming from 60 W at the same distance is $\sqrt{\frac{x}{5}}$ E, where the value of x = _____

25. The first three spectral lines of H-atom in the Balmer series are given $\lambda_1, \lambda_2, \lambda_3$ considering the Bohr atomic model, the wavelengths of first and third spectral lines $\left(\frac{\lambda_1}{\lambda_3}\right)$ are related by a factor of approximately $'p' \times 10^{-1}$. The value of p, to the nearest integer, is _____.

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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. At 300 K, 150 g of 80% (w/w) aqueous H_2SO_4 solution is mixed with 250 g of 20% (w/w) aqueous H_2SO_4 solution. What is the mole fraction of H_2SO_4 in the resultant solution? (Given: Atomic mass

H = 1 u; S = 32 u; O = 16 u)

A) 0.12 B) 0.186 C) 0.234 D) 0.356

27. Consider the following reaction sequence:

$$But-2-yne \xrightarrow[(ii)H_2O]{(i)Hg^{+2}/H_2SO_4} A \xrightarrow{CH_3MgBr} B$$

$$\xrightarrow{Conc.H_2SO_4.\Delta} C \xrightarrow[(ii)Zn/H_2O]{(i)O_3} D+E$$

The products D and E are:

A) Propanal and Methanal B) Acetaldehyde and Acetaldehyde

C) Propanone and Ethanal D) Acetone and Formaldehyde

28. 4.8 g of a metal 'M' (Molar mass: 24 g/mol) reacts completely with 3.2 g of oxygen to form its oxide. Which of the following statements are correct?

1) The formula of the oxide is MO.

2) The equivalent weight of the metal is 12 g/eq.

3) 6.0 g of the same metal would require 4.0 g of oxygen for complete reaction.

4) The volume of oxygen at STP consumed in the reaction with 4.8 g of metal is 2.24 L.

Choose the correct answer from the options given below:

A) 1, 2 and 3 only B) 1, 2 and 4 only C) 2, 3 and 4 only D) 1, 2, 3, 4

29. Given below are two statements:

Statement-I: The first ionization enthalpy of K is lower than that of Ca.

Statement-II: The second ionization enthalpy of Ca is lower than that of K.

In the light of the above statements, choose the correct answer from the options given below:

A) Both Statement-I and Statement-II are false.

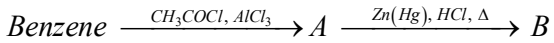
B) Statement-I is true but Statement-II is false.

C) Both Statement-I and Statement-II are true.

D) Statement-I is false but Statement-II is true.

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30. The final product [B] in the following reaction sequence is:



A) Toluene B) Ethylbenzene C) Cumene D) Acetophenone

31. In Dumas' method for estimation of nitrogen, 0.25 g of an organic compound gave 40 mL of nitrogen collected at 300 K and 740 mm pressure. The percentage of nitrogen in the compound is approximately:

(Given: Aqueous tension at 300 K is 24 mm)

A) 14.5 B) 17.14 C) 18.7 D) 20.3

32. Correct statements regarding chemical kinetics among the following are:

1) For a zero-order reaction, the rate constant has the same units as the rate of reaction.

2) The half-life of a first-order reaction is independent of initial concentration.

3) A catalyst increases the rate of reaction by providing an alternative path with lower activation energy.

4) The rate constant of a reaction decreases with increase in temperature.

Choose the correct answer from the options given below:

A) 1, 2 and 3 only B) 1 and 2 only

C) 1, 2 and 4 only D) 2, 3 and 4 only

33. The IUPAC name of the following compound is:



A) 3-Hydroxy-4-methylhexanal

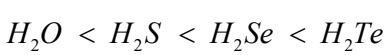
B) 4-Hydroxy-3-methylhexanal

C) 3-Hydroxy-4-methylpentanal

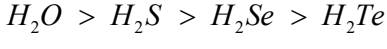
D) 4-Hydroxy-3-methylpentanal

34. Given below are two statements:

Statement-I: The acidic strength of hydrides of group 16 follows the order:



Statement-II: The thermal stability of hydrides of group 16 follows the order:



In the light of the above statements, choose the correct answer from the options given below:

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A) Both Statement-I and Statement-II are true.

B) Statement-I is true but Statement-II is false.

C) Both Statement-I and Statement-II are false.

D) Statement-I is false but Statement-II is true.

35. Match List-1 with List-II.

List-1 (Reaction)	**List-II (Product)**
P) Glucose with Tollen's reagent	I) Glucose oxime
Q) Glucose with acetic anhydride	II) Saccharic acid
R) Glucose with hydroxylamine	III) Gluconic acid
S) Glucose with conc. HNO_3	IV) Glucose pentaacetate

Choose the correct answer from the options given below:

A) P-III, Q-IV, R-I, S-II B) P-I, Q-IV, R-III, S-II

C) P-III, Q-I, R-IV, S-II D) P-II, Q-IV, R-I, S-III

36. Among NH_3, PH_3, AsH_3, SbH_3, and H_2O, identify the molecule (X) with the highest dipole moment. The number of lone pairs of electrons present on the central atom of the molecule (X) is:

A) 2 B) 0 C) 1 D) 3

37. Given below are two statements:

Statement-I: The correct order of first ionization enthalpy for the elements Li, Be, B, C is Li < B < Be < C.

Statement-II: The correct order of electron gain enthalpy for the elements F, Cl, Br, I is Cl > F > Br > I.

In the light of the above statements, choose the correct answer from the options given below:

A) Statement-I is false but Statement-II is true

B) Both Statement-I and Statement-II are true

C) Both Statement-I and Statement-II are false

D) Statement-I is true but Statement-II is false

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38. Which of the following mixtures gives a buffer solution with pH = 9.26?

Given: pKb of NH_4OH = 4.74

A) 0.1 M NH_4OH (0.5 L) + 0.1 M HCl (0.25 L)

B) 0.2 M NH_4OH (0.5 L) + 0.1 M HCl (0.25 L)

C) 0.1 M NH_4OH (0.5 L) + 0.1 M HCl (0.5 L)

D) 0.3 M NH_4OH (0.5 L) + 0.1 M HCl (0.5 L)

39. The energy of the first line of the Lyman series of a hydrogen-like ion is 40.8 eV. The energy (in eV) of the first line of the Balmer series of this ion is:

A) 40.8 B) 20.4 C) 7.56 D) 10.2

40. Identify the correct statements about secondary standards:

1. They are less pure than primary standards.
2. Their solutions must be standardized against a primary standard.
3. They are always hygroscopic in nature.
4. They have high molar mass to minimize weighing errors.
5. They can be used directly to prepare standard solutions of accurate concentration.

Choose the correct answer from the options given below:

A) 1 and 2 only B) 1, 2 and 3 only C) 2, 4 and 5 only D) 1, 3 and 5 only

41. $\left[Ni(CN)_4\right]^{-2}$ is a diamagnetic complex. Identify the INCORRECT statements about this complex.

1. The complex exhibits geometrical isomerism.
2. The complex is yellow in color.
3. The calculated spin-only magnetic moment of the complex is 0 BM.
4. The hybridization of Ni in this complex is sp^3.
5. The complex is an outer orbital complex.

Choose the correct answer from the options given below:

A) 1 and 2 only B) 1, 4 and 5 only C) 3 and 4 only D) 2, 3 and 5 only

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42. Consider the following reduction processes:

$Ag^{+} + e^{-} \rightarrow Ag(s),\ E^{o} = +0.80\ V$

$Cu^{2+} + 2e^{-} \rightarrow Cu(s),\ E^{o} = +0.34\ V$

$Fe^{2+} + 2e^{-} \rightarrow Fe(s),\ E^{o} = -0.44\ V$

$Zn^{2+} + 2e^{-} \rightarrow Zn(s),\ E^{o} = -0.76\ V$

The tendency to act as a reducing agent decreases in the order:

A) Zn > Fe > Cu > Ag

B) Ag > Cu > Fe > Zn

C) Zn > Fe > Ag > Cu

D) Fe > Zn > Cu > Ag

43. The compound A, C_8H_8O, reacts with acetophenone to form a single product via cross-aldol condensation. Compound A on reaction with Fehling's solution gives a reddish-brown precipitate. Compound A is:

A) Phenylacetaldehyde

B) 4-Methylbenzaldehyde

C) 2-Methylbenzaldehyde

D) Acetophenone

44. 2-Methylbutan-2-ol cannot be prepared by:

1) $CH_3 - MgBr + CH_3COCH_2CH_3$

2) $CH_3 - CH_2 - MgBr + CH_3COCH_3$

3) $(CH_3)_2 CH - CH_2 - Cl + aq.KOH$

4) $(CH_3)_2 C = CH_2 + H_2O / H^{+}$

5) $CH_3 - MgBr + HCOOC_2H_5$

Choose the correct answer from the options given below:

A) 1 and 5 only B) 3 and 5 only C) 2 and 3 only D) 3, 4 and 5 only

45. The dibromo compound [P] (molecular formula: $C_9H_{10}Br_2$) when heated with excess sodamide followed by treatment with dilute HCl gives [Q]. On warming [Q] with mercuric sulphate and dilute sulfuric acid yields [R] which gives positive 2,4 - DNP test but negative Tollen's test. The compound [P] is:

A) 1,2-Dibromo-1-phenylpropane

B) 2,3-Dibromo-2-phenylbutane

C) 1,2-Dibromo-1-phenyl-2-methylpropane

D) 1,4-Dibromo-2-phenylbutane

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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46. Given: $Ag(s) \mid AgBr(s) \mid Br^{-}\ (1.0\ M) \parallel Cl^{-}\ (0.5\ M) \mid AgCl(s) \mid Ag(s)$

If the emf of following cell is 0.12 V then the ratio of $K_{sp}(AgCl)$ *and* $K_{sp}(AgBr)$ is _____.

[Given: $2.303\ \frac{RT}{F} = 0.06$]

47. Using the following energy values, determine the lattice energy of KF(s):

Property	Energy ($kJ\ mol^{-1}$)
$\Delta H_{atomization}\ of\ K(s)$	90
$\Delta H_{atomization}\ of\ F_2(s)$	158
$\Delta H_f\ of\ KF(s)$	-567
IE of K(g)	419
EA of F(g)	328

48. Consider the following oxides of d-block elements:

$V_2O_5, Cr_2O_3, Mn_2O_7, V_2O_3, V_2O_4$

Number of oxides which are acidic is x. Consider the following complex compound $\left[Co(NH_2CH_2CH_2NH_2)_3\right]_2(SO_4)_3$, the primary valency of complex is y. The value of (x + y) is:

49. The mass of acetanilide obtained from the acetylation reaction of 4.65 g of aniline, if the yield of the product is 80%, is_____ g. (Nearest integer)

(Given molar mass in g/mol: H:1, C:12, N:14, O:16)

50. Consider $A \xrightarrow{k_1} B$ and $C \xrightarrow{k_1} D$ are two reactions.

If the rate constant (k_1) of the A → B reaction can be expressed by the following equation

$$\log_{10} k = 12.34 - \frac{2.0 \times 10^4}{T/K}$$

and activation energy of C → D reaction (E_{a_2}) is $\frac{1}{4}$th of the A → B reaction (E_{a_1}), then the value of E_{a_2} is _____ $kJ\ mol^{-1}$. (Nearest Integer)

Given: R = 8.314 J $mol^{-1}K^{-1}$

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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Let " n " be the number obtained on rolling a fair die. If the probability that the system of equations $x-ny+z=10$, $x+(n-3)y+(n+2)z=12$, $(n-2)y+z=2$ has a unique solution is " $\frac{k}{6}$ " then the sum of " k " and all possible values of " n " is

A) 18 B) 22 C) 25 D) 27

52. If the mean deviation about the median of the numbers k, 2k, 3k, ... , 500k is 250, then k^2 is equal to:

A) 4 B) 16 C) 9 D) 1

53. Find the number of elements in the relation $R=\{(x,y):x^2+9y^2\le 40, x,y\in Z\}$

A) 65 B) 45 C) 50 D) 35

54. Let $S=\{z\in C:2z^2+\bar{z}=0\}$. Find the value of $\sum_{z\in S}|z|^2$

A) 3/4 B) 1 C) 2 D) 1/2

55. If $\lim_{x\to 0}\frac{a\sin x+b\tan x+cx^2}{x^3}$=2. Find the value of a+b+c

A) 1 B) 2 C) 3 D) 0

56. If y = y(x) satisfies the differential equation $\frac{dy}{dx}=\frac{1+y^2}{2\sqrt{x}(1+x)}$ with the initial condition y(1) = 0, find y(3).

A) $2+\sqrt{3}$ B) $2-\sqrt{3}$ C) $4-\sqrt{3}$ D) $4+\sqrt{3}$

57. **Among the statements:**

(S1): If A(2,3) and B(4, -1) are two vertices of a triangle whose orthocentre is (0, 0), then the third vertex C lies on the line x + 2y = 0.

(S2): If positive numbers a, b, c are in G.P., then the lines ax + by + c = 0 are always tangent to the parabola $y^2=4x$

A) both are true B) S1 is true S2 is false

C) S1 is false S2 is true D) both are false

58. Let a, b, c be vectors such that $\bar{c}=\bar{a}\times\bar{b}, if\ |a|=2, |b|=5$ and the angle between them is $\frac{\pi}{6}$ find the maximum value of $(\bar{c}.\bar{d})^2$ for any unit vector $\bar{d}$

A) 25 B) 35 C) 50 D) 45

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59. Find the value of $x+y+z$. *If* $AX=B$ *with* $B=[4,2,2]^T$, $X=[x,y,z]^T$, $|A|=-2$, and adj $A=\begin{bmatrix}1 & -2 & 1\\ 2 & 0 & 1\\ 1 & 2 & 1\end{bmatrix}$

A) -3 B) -11 C) -5 D) 2

60. Let L_1 be the line $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{2}$ and L_2 be the line $\frac{x-1}{0}=\frac{y-2}{1}=\frac{z-3}{0}$. Let S be the set of all points (a, b, c) on L_1 whose distance from the point of intersection of L_1 *and* L_2 is 6 units. Then $\sum(a+b+c)$ is

A) 12 B) 24 C) 18 D) 30

61. Let P(6, 3) be a point on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. If the eccentricity of the hyperbola is $\sqrt{2}$ (rectangular hyperbola), then the area of $\Delta PSS'$ is

A) $18\sqrt{2}$ B) 36 C) 18 D) $9\sqrt{6}$

62. The area of the region $A=\{(x,y):x^2+y^2\le 4 \text{ and } y^2\le 3x\}$ is

A) $\frac{4\pi}{3}+\frac{1}{\sqrt{3}}$ B) $\frac{4\pi}{3}+\frac{2}{\sqrt{3}}$ C) $\frac{2\pi}{3}+\frac{\sqrt{3}}{2}$ D) $\frac{2\pi}{3}+\frac{2}{\sqrt{3}}$

63. Let α,β be the roots of the quadratic equation $x^2-10x+\lambda=0$, where $\lambda\in Z$. *If* $2\le|\beta-\alpha|\le 6$, then the sum of all possible values of λ is

A) 60 B) 66 C) 180 D) 54

64. Let the domain of $f(x)=\log_2(x^2-5x+6)+\sin^{-1}\left(\frac{|2x-1|}{5}\right)$ be $(\beta,\gamma]$, Then $\gamma-\beta$ is

A) 5 B) 8 C) 4 D) 6

65. Let $f(x)=\min\{|x|,1-x^2\}$, If S is the set of points where $f(x)$ is not differentiable, then the sum of elements in S is

A) 0 B) 1 C) -1 D) 2

66. Let S and S' be the foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$. If P is a point on the ellipse such that the product of its focal distances SP . S'P = 20, then the distance of P from the center of the ellipse is

A) 17 B) 19 C) 21 D) 23

67. Let S be the locus of the midpoint of chords of $y^2=4ax$ which pass through (0, 0). If Q is a point that divides the segment from the origin to a point P on S in the ratio 1 : 3 (internally), then the locus of Q is:

A) $4y^2=ax$ B) $y^2=ax$ C) $2y^2=ax$ D) $y^2=4ax$

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68. Let $f(x)=[x]^2-2[x]-3$. Which of the following is true

A) f(x) = 0 has exactly two solutions.

B) f(x) = 0 has infinitely many solutions in the interval $[-1,0)\cup[3,4)$

C) f(x)>0 for all x>3.

D) $\int_0^2 f(x)\,dx=-6$

69. Let f and g be functions satisfying $f(x+y)=f(x)f(y)$ *with* $f(1)=3$ and $g(x+y)=g(xy)$, with $g(1)=2$, $g(x)$ is a polynomial function for all $x,y\in N$. *If* $\sum_{x=1}^{n}\frac{f(x)}{g(x)}=546$, then n is equal to

A) 4 B) 5 C) 6 D) 10

70. Let $P_n=\sum_{r=0}^{n}\frac{(-1)^r\,C_r}{r+1}$. The value of $\sum_{n=1}^{50}\frac{1}{P_n}$ is

A) 1275 B) 1325 C) 2550 D) 50

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Let $a=\hat{i}+\sqrt{2}\hat{j}+\lambda\hat{k}$ where $\lambda>0$. Suppose $\vec{a}$ makes an acute angle with $b=2\hat{i}+\sqrt{2}\hat{j}-\lambda^2\hat{k}$. If the angle ϕ that vector $\vec{a}$ makes with the positive y-axis satisfies $\cos\phi>\frac{1}{2}$. Then number of possible integer values for λ.

72. Let $\alpha=\int_0^{16}\left(\left[x^{1/2}\right]\right)dx$. Find the value of $\frac{1}{\pi}\int_0^{\alpha\pi}\sin^2\theta\,d\theta$

73. Let $\sin(\alpha+\beta)=-\frac{4}{5}$ and $\cos(\alpha-\beta)=-\frac{12}{13}$ if $(\alpha+\beta)\in\left(\pi,\frac{3\pi}{2}\right)$ and $(\alpha-\beta)\in\left(\frac{\pi}{2},\pi\right)$, and $\tan(2\alpha)=\frac{p}{q}$ where p,q are coprime integers, find $p+q$

74. Suppose a,b,c are in A.P and $a^2,3b^2,c^2$ are in G.P . If $a<b<c$ and $a+b+c=3$, Find the value of $a^2+b^2+c^2$

75. Let $A=\{1,2.3.....10\}$. Find the number of subsets B of A such that $B\cap\{1,2,3\}\neq\phi$ and the smallest element in B is even

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